

SAFETY DATA SHEET

BRSBENT SQ — Premium Bentonite (Montmorillonite)

Prepared in accordance with the Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

Section 1: Identification

Product identifier: BRSBENT SQ

Chemical name: Bentonite (sodium montmorillonite clay), HDD drilling grade

Recommended use: Base material of HDD drilling fluids — viscosifier and filtration-control material for cooling, lubrication, cuttings carrying, and borehole wall protection

Manufacturer / Supplier: Brighter Star Drilling Fluids

Website: www.pcenigeria.com

Section 2: Hazards Identification

Hazard classification: Not classified as a hazardous substance under GHS / OSHA Hazard Communication Standards. The following minor hazards exist in practice

- Dust irritation — inhalation, skin contact, or eye contact with bentonite dust may cause mild irritation to the respiratory tract, skin, and eyes; ingestion may irritate the digestive tract
- Non-combustible mineral — the product is an inorganic clay and does not burn; no fire or dust-explosion hazard
- Slip hazard — spilled powder, or powder wetted with water, forms an extremely slippery film on floors and work surfaces
- Quartz content — the product contains 1–2% naturally occurring quartz; avoid prolonged inhalation of dust

Section 3: Composition / Information on Ingredients

Chemical characterization: Naturally occurring mineral (mixture).

Component	CAS No.
Montmorillonite (content 85–90%)	1318-93-0
Quartz (content 1–2%)	7631-86-9
Functional additives, trade secret (content 6–8%)	—

Section 4: First-Aid Measures

Exposure Route	Action
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Eye contact	Rinse immediately with plenty of clean water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Remove contact lenses if present. Seek medical attention if irritation persists
Skin contact	Remove contaminated clothing and shoes. Wash the affected area thoroughly with soap and plenty of water. Launder clothing before reuse. Seek medical advice if irritation develops or persists
Inhalation	Move the victim to fresh air immediately and keep at rest in a comfortable breathing position. If breathing is difficult, administer oxygen; if not breathing, give artificial respiration. Seek medical attention
Ingestion	Do NOT induce vomiting. Rinse mouth thoroughly with water. If conscious and alert, drink 2–4 cupfuls of water or milk. Never give anything by mouth to an unconscious person. Seek medical attention

Note to physician: Treat symptomatically and supportively. No specific antidote is known.

Section 5: Fire-Fighting Measures

Suitable extinguishing media: The product itself does not burn. Use extinguishing measures appropriate to local circumstances and the surrounding environment (water spray, dry chemical, carbon dioxide, or foam)

Unsuitable media: Not applicable — the product is non-combustible

Autoignition temperature: Not applicable — the product itself does not burn

NFPA rating: Health 0 · Flammability 0 · Instability 0

Hazardous combustion products: None known — the product is an inorganic mineral and does not undergo hazardous thermal decomposition

Protection of firefighters: Wear self-contained breathing apparatus (SCBA, MSHA/NIOSH-approved or equivalent) in pressure-demand mode and full protective gear. Fight fire from upwind

Precaution: Product wetted during fire-fighting forms a slippery slurry — take care to avoid slip hazards

Section 6: Accidental Release Measures

- Isolate the spill area and keep unauthorized personnel away
- Ventilate the area and wear appropriate PPE (dust mask, gloves, goggles, protective clothing)
- Avoid generating dust — dry sweeping is not recommended. Use a vacuum with a HEPA filter to collect the powder, or pre-dampen with water before sweeping
- Collect the material into clean, sealed containers for recycling or disposal
- Prevent the product from entering drains, sewers, surface water, or groundwater
- Clean the affected surface with water as appropriate — caution: wet bentonite forms an extremely slippery film
- Dispose of contaminated material in accordance with local environmental regulations

Section 7: Handling and Storage

Handling: Avoid dust formation at all times and provide adequate local exhaust ventilation where dust may be generated. Do not breathe dust; avoid contact with eyes, skin, and clothing. Handle gently using scoops, funnels, or spoons; avoid shaking or vigorous stirring that disperses dust. Prohibit eating, drinking, and smoking in handling areas; wash hands thoroughly with soap and water after handling. Keep containers tightly closed when not in use. Access limited to trained personnel only

Storage: Store in a cool, dry, clean, and well-ventilated place away from direct sunlight, at 5–40 °C and relative humidity ≤ 80% (ideally < 50%). Bentonite is hygroscopic — moisture ingress causes caking and loss of swelling performance. Keep containers tightly closed; after opening, reseal the inner PE bag immediately. No smoking in storage areas. Stack bags on pallets off the ground and do not stack excessively high. Do not store together with toxic, harmful, or odor-emitting substances; keep away from food, feed, and drinking water

Shelf life: 24 months from the date of manufacture under proper sealed-storage conditions

Section 8: Exposure Controls / Personal Protection

Protection	Recommended Measures
Eyes	Chemical safety goggles compliant with OSHA 29 CFR 1910.133 or EN 166; full goggles recommended in dusty environments
Skin / hands	Impervious rubber gloves (nitrile or latex preferred); protective clothing or aprons; long sleeves
Respiratory	NIOSH/MSHA-approved dust mask or respirator (EN 149 P1/P2 or equivalent) when dust levels are significant or ventilation is inadequate
Footwear	Safety shoes
Facilities	Emergency eyewash fountain and safety shower must be accessible in the immediate work area

Hygiene measures: Wash hands thoroughly with soap and water after handling and before eating or drinking. Launder contaminated clothing before reuse.

Section 9: Physical and Chemical Properties

Property	Value
Appearance	Greyish-white free-flowing powder
Odor	Odorless
Taste	Tasteless
Solubility	Insoluble in water — swells and forms a colloidal suspension; insoluble in ethanol, ether, acetone, and other organic solvents
Autoignition temperature	Not applicable — the product itself does not burn

Property	Value
Hygroscopicity	Hygroscopic — cakes on moisture ingress; swelling performance decreases

Section 10: Stability and Reactivity

Chemical stability: Stable under recommended storage conditions

Incompatible materials: No known incompatible materials

Conditions to avoid: Dust formation and moisture ingress

Hazardous decomposition products: None known — the product is thermally stable; in the event of fire see Section 5

Hazardous reactions: No known hazardous reactions under normal conditions

Section 11: Toxicological Information

Acute toxicity: Non-toxic; not classified

Irritation: Dust may cause mild irritation to the respiratory tract, skin, and eyes; ingestion may irritate the digestive tract

Medical treatment: Treat symptomatically and supportively; no specific antidote is known

Section 12: Ecological Information

Biodegradability: Not applicable — inorganic mineral; not biodegradable

Environmental precautions: Prevent the product from entering drains, sewers, surface water, or groundwater; spills must be contained and properly disposed of

Section 13: Disposal Considerations

- Dispose of the product and its container in accordance with local, regional, and national regulations
- Uncontaminated residual product may be landfilled where permitted
- Do NOT discharge bentonite powder or slurry into drains, sewers, or waterways — it swells and may block pipework
- Empty containers should be cleaned of residual powder before disposal or recycling

Section 14: Transport Information

Not classified as a hazardous substance; not regulated as a dangerous good for transport.

Section 15: Regulatory Information

GHS / OSHA HCS: Not classified as a hazardous substance

NFPA 704: Health 0 · Flammability 0 · Instability 0

Section 16: Other Information

Remark: The above safety precautions are compiled from industry-standard SDS references for Bentonite. For formal export documentation, regulatory submission, or batch-specific compliance, always refer to the official SDS issued by the manufacturer for the exact grade and batch — parameters such as montmorillonite content, swelling index, and impurity profile may vary between grades and manufacturers.

Disclaimer: The information contained in this Safety Data Sheet is provided for reference only and is based on the current state of knowledge. It describes the product with regard to safety requirements and does not guarantee any specific properties. In all cases, the actual conditions of use shall prevail. To the fullest extent permitted by applicable law, Brighter Star Drilling Fluids assumes no responsibility or liability for any results, losses, or damages arising from the use of, or reliance on, this document.